

Partial Differentiation

$$z = f(x, y)$$

$$\frac{dz}{dx} = \lim_{dx \rightarrow 0} \frac{f(x+dx, y) - f(x, y)}{dx}$$

$$\frac{dz}{dy} = \lim_{dy \rightarrow 0} \frac{f(x, y+dy) - f(x, y)}{dy}$$

$$\frac{d}{dx} \left(\frac{dz}{dx} \right) = \frac{d^2 z}{dx^2} = \frac{d^2 f}{dx^2}$$

$$\frac{d^2 z}{dy dx} = \frac{d^2 z}{dx dy}$$

Q. $z = x^3 + y^3 - 3axy$ prove that $\frac{d^2 z}{dx dy} = \frac{d^2 z}{dy dx}$

$$\rightarrow \frac{dz}{dx} = 3x^2 + 0 - 3ay \quad \frac{dz}{dy} = 0 + 3y^2 - 3ax$$

$$\frac{d}{dy} \left(\frac{dz}{dx} \right) = 0 + 0 - 3a$$

$$\frac{d}{dx} \left(\frac{dz}{dy} \right) = 0 - 3a$$

$$\frac{d^2 z}{dy dx} = -3a$$

$$\frac{d^2 z}{dx dy} = -3a$$

hence proved.

Q. $z(x+y) = x^2 + y^2$ prove that $\left(\frac{dz}{dx} - \frac{dz}{dy} \right)^2 = 4 \left(1 - \frac{dz}{dx} - \frac{dz}{dy} \right)$

$$\Rightarrow z = \frac{x^2 + y^2}{x+y}$$

$$\frac{dz}{dx} = \frac{(x+y)(2x) - (x^2 + y^2)(1)}{(x+y)^2} = \frac{2x^2 + 2xy - x^2 - y^2}{(x+y)^2}$$

$$\Rightarrow \frac{dz}{dx} = \frac{x^2 + 2xy - y^2}{(x+y)^2}$$

$$\frac{dz}{dy} = \frac{(x+y)(2y) - (x^2 + y^2)(1)}{(x+y)^2} = \frac{y^2 - x^2 + 2xy}{(x+y)^2}$$

$$\left(\frac{dz}{dx} - \frac{dz}{dy}\right)^2 = 4\left(1 - \frac{dz}{dx} - \frac{dz}{dy}\right)$$

$$\left[\frac{x^2 - y^2 + 2xy - y^2 + x^2 - 2xy}{(x+y)^2}\right]^2 = 4\left[1 - \frac{(x^2 - y^2 + 2xy + y^2 - x^2 + 2xy)}{(x+y)^2}\right]$$

$$\frac{(2x^2 - 2y^2)^2}{(x+y)^4} = 4 \frac{(x^2 + y^2 - 4xy)}{(x+y)^2}$$

$$4 \frac{(x^2 - y^2)^2}{(x+y)^2} = 4(x^2 + (x-y)^2 - 2xy)$$

$$\frac{(x-y)^2(x-y)^2}{(x+y)^2} = (x-y)^2$$

$$(x-y)^2 = (x-y)^2$$

hence proved

* Homogeneous function

$$f(x, y) = a_0 x^n y^0 + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \dots + a_n x^0 y^n$$
$$= x^n \left[a_0 + a_1 \frac{y}{x} + \dots + a_n \frac{y^n}{x^n} \right]$$

$$f(x, y) = x^n F\left(\frac{y}{x}\right)$$

$$f(x, y) = y^n F\left(\frac{x}{y}\right)$$

Q. Euler's Theorem on Homogeneous function

Theorem-1

If $f(x, y)$ is a homogeneous function of x and y of degree n then

$$x \frac{df}{dx} + y \frac{df}{dy} = nf$$

proof - Since f is a homogeneous function of x & y of degree n we can write

$$f(x, y) = x^n F\left(\frac{y}{x}\right)$$

$$\frac{df}{dx} = x^n F'\left(\frac{y}{x}\right) \left(-\frac{y}{x^2}\right) + F\left(\frac{y}{x}\right) n x^{n-1}$$

$$\frac{df}{dx} = \frac{1}{x} \left[-x^{n-1} F'\left(\frac{y}{x}\right) y + n x^n F\left(\frac{y}{x}\right) \right]$$

$$x \frac{df}{dx} = \left[n x^n F\left(\frac{y}{x}\right) - x^{n-1} F'\left(\frac{y}{x}\right) y \right] \quad \text{--- (1)}$$

$$\frac{df}{dx} = x^n F'\left(\frac{y}{x}\right) \left(-\frac{y}{x}\right) \quad \therefore f = x^n F\left(\frac{y}{x}\right)$$

$$y \frac{df}{dy} = x^n F\left(\frac{y}{x}\right) \left(\frac{y}{x}\right) \quad \text{--- (2)}$$

$$\begin{aligned}
 \Rightarrow x \frac{dF}{dx} + y \frac{dF}{dy} &= nF \\
 &= nx^{n-1} F\left(\frac{y}{x}\right) - x^{n-1} F'\left(\frac{y}{x}\right) y + x^n F'\left(\frac{y}{x}\right) \left(\frac{y}{x}\right) = nF \\
 &= x^n \left[n F\left(\frac{y}{x}\right) - x F'\left(\frac{y}{x}\right) \left(\frac{y}{x}\right) + F'\left(\frac{y}{x}\right) \left(\frac{y}{x}\right) \right] = nF \\
 &= x^n n F\left(\frac{y}{x}\right) = nF \\
 &= nF = nF
 \end{aligned}$$

Theorem-2

If u is a homogeneous function of degree n , then

$$i) x \frac{d^2 u}{dx^2} + y \frac{d^2 u}{dx dy} = (n-1) \frac{du}{dx}$$

$$ii) x \frac{d^2 u}{dx dy} + y \frac{d^2 u}{dy^2} = (n-1) \frac{du}{dy}$$

$$iii) x^2 \frac{d^2 u}{dx^2} + 2xy \frac{d^2 u}{dx dy} + y^2 \frac{d^2 u}{dy^2} = n(n-1)u$$

proof - Since u is a homogeneous function of x and y of degree n , therefore by Euler's theorem, we have

$$x \frac{du}{dx} + y \frac{du}{dy} = nu \quad (1)$$

$$① x \frac{d^2 u}{dx^2} + \frac{du}{dx} + y \frac{d^2 u}{dy dx} = n \frac{du}{dx}$$

$$x \frac{d^2 u}{dx^2} + y \frac{d^2 u}{dy dx} = (n-1) \frac{du}{dx} \quad (2)$$

$$② x \frac{d^2 u}{dx dy} + y \frac{d^2 u}{dy^2} + \frac{du}{dy} = n \frac{du}{dy}$$

$$x \frac{d^2u}{dx dy} + y \frac{d^2u}{dy^2} = (n-1) \frac{du}{dy} \quad \text{--- (3)}$$

③ in eqn (2) $\times x$ + in eqn (3) $\times y$

$$\begin{aligned} x^2 \frac{d^2u}{dx^2} + xy \frac{d^2u}{dx dy} &= x(n-1) \frac{du}{dx} \\ + xy \frac{d^2u}{dx dy} + y^2 \frac{d^2u}{dy^2} &= y(n-1) \frac{du}{dy} \end{aligned}$$

$$x^2 \frac{d^2u}{dx^2} + y^2 \frac{d^2u}{dy^2} + 2xy \frac{d^2u}{dx dy} = (n-1) \left(x \frac{du}{dx} + y \frac{du}{dy} \right)$$

$$\therefore x \frac{dF}{dx} + y \frac{dF}{dy} = nF$$

$$x \frac{du}{dx} + y \frac{du}{dy} = nu$$

$$\Rightarrow x^2 \frac{d^2u}{dx^2} + y^2 \frac{d^2u}{dy^2} + 2xy \frac{d^2u}{dx dy} = n(n-1)u$$

Q $u = \sin^{-1} \left(\frac{x^2 + y^2}{x+y} \right)$

$\rightarrow x \frac{du}{dx} + y \frac{du}{dy} = \tan u$ (To prove)

$$\Rightarrow \sin u = \frac{x^2 + y^2}{x+y}$$

$x \rightarrow xt, y \rightarrow yt$

$$\sin u = \frac{x^2 t^2 + y^2 t^2}{xt + yt} = \frac{t(x^2 + y^2)}{x+y}$$

$$n \equiv 1$$

$$\frac{du}{dx} = \frac{d \sin u}{dx} = \cos u \frac{du}{dx}$$

$$\frac{du}{dy} = \frac{d \sin u}{dy} = \cos u \frac{du}{dy}$$

$$x \frac{du}{dx} + y \frac{du}{dy} = nu$$

$$x \cos u \frac{du}{dx} + y \cos u \frac{du}{dy} = 1 \sin u$$

$$x \frac{du}{dx} + y \frac{du}{dy} = \tan u$$

hence proved.

Q $u = \sin^{-1} \frac{x^2 y^2}{x+y}$

To prove $x \frac{du}{dx} + y \frac{du}{dy} = 3 \tan u$

$$\Rightarrow x \frac{du}{dx} + y \frac{du}{dy} = nu$$

$$\Rightarrow \sin u = \frac{x^2 y^2}{x+y}$$

$$x \rightarrow xt, y \rightarrow yt$$

$$\sin u = \frac{x^2 t^2 y^2 t^2}{xt + yt} = t^3 \frac{x^2 y^2}{x+y}$$

$$n=3$$

$$\frac{du}{dx} = \cos u \frac{du}{dx} \quad \rightarrow \quad \frac{du}{dy} = \cos u \frac{du}{dy}$$

$$x \cos u \frac{du}{dx} + y \cos u \frac{du}{dy} = 3 \sin u$$

$$x \frac{du}{dx} + y \frac{du}{dy} = 3 \tan u \quad \text{hence proved}$$

Q. $u = \log \left(\frac{x^4 + y^4}{x+y} \right)$

$\Rightarrow \frac{x du}{dx} + y \frac{du}{dy} = 3$ (To prove)

$e^u = \frac{x^4 + y^4}{x+y}$

$x \rightarrow xt, y \rightarrow yt$

$e^u = \frac{t^4 (x^4 + y^4)}{t(x+y)} = t^3 \frac{(x^4 + y^4)}{x+y}$

$n=3$

$\frac{du}{dx} = \frac{d e^u}{dx} = e^u \frac{du}{dx} ; \frac{du}{dy} = \frac{d e^u}{dy} = e^u \frac{du}{dy}$

$\therefore x \frac{du}{dx} + y \frac{du}{dy} = nu$

$\rightarrow x e^u \frac{du}{dx} + y e^u \frac{du}{dy} = 3 e^u$

$\frac{x du}{dx} + y \frac{du}{dy} = 3$

hence proved

Q. $u = \frac{xy}{x+y}$ show that $x^2 \frac{d^2 u}{dx^2} + 2xy \frac{d^2 u}{dx dy} + y^2 \frac{d^2 u}{dy^2} = 0$

$\Rightarrow u = \frac{xy}{x+y} \Rightarrow x \rightarrow xt, y \rightarrow yt$

$\Rightarrow u = \frac{t^2(xy)}{t(x+y)} = t \frac{xy}{x+y}$

$n=1$

$x^2 \frac{d^2 u}{dx^2} + 2xy \frac{d^2 u}{dx dy} + y^2 \frac{d^2 u}{dy^2} = (n-1)u$

$\boxed{x^2 \frac{d^2 u}{dx^2} + 2xy \frac{d^2 u}{dx dy} + y^2 \frac{d^2 u}{dy^2} = 0} \quad (\because n=1)$

★ formula

$$i) \frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt}$$

$$ii) \frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx}$$

$$iii) \frac{dy}{dx} = \frac{\partial f / \partial x}{\partial f / \partial y}$$

ex:- $x^y + y^x = c$, find $\frac{dy}{dx}$

$$\frac{\partial f}{\partial x} = yx^{y-1} + y^x \log y \quad [\because \frac{da^x}{dx} = a^x \log a]$$

$$\frac{\partial f}{\partial y} = x^y \log x + xy^{x-1}$$

$$\frac{dy}{dx} = \frac{\partial f / \partial x}{\partial f / \partial y} = \frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$$

ex:- $u = x \log xy$, where

$$x^3 + y^3 + 3xy = 1 \text{ find } \frac{du}{dx}$$

$$\Rightarrow \frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx}$$

$$\frac{\partial u}{\partial x} = \log xy + \frac{x}{xy} y \Rightarrow \log xy + 1$$

$$\frac{\partial u}{\partial y} = \frac{x}{xy} x = \frac{x}{y}$$

$$\frac{dy}{dx} = ?$$

$$3x^2 + 3y^2 \frac{dy}{dx} + 3y + 3x \frac{dy}{dx} = 0$$

$$(3y^2 + 3x) \frac{dy}{dx} = -3x^2 - 3y$$

$$\frac{dy}{dx} = - \frac{(x^2 + y)}{y^2 + x}$$

$$\Rightarrow \frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx}$$

$$\frac{du}{dx} = \log xy + 1 + \frac{x}{y} \left(- \frac{(x^2 + y)}{(y^2 + x)} \right) \quad \text{Ans.}$$

Jacobian

$$u = f(x, y)$$

$$v = f(x, y)$$

$$\frac{\partial(u, v)}{\partial(x, y)} = J(u, v) = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{vmatrix}$$

$$\begin{array}{l} u = f(x, y, z) \\ v = f(x, y, z) \\ w = f(x, y, z) \end{array} \quad \begin{vmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{vmatrix}$$

$$i) \frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(u, v)} = 1$$

$$JJ' = 1$$

$$ii) \frac{\partial(u, v)}{\partial(t_1, t_2)} = \frac{\partial(u, v)}{\partial(x, y)} \frac{\partial(x, y)}{\partial(t_1, t_2)}$$

Q. $x = r \cos \theta$, $y = r \sin \theta$ find $\frac{\partial(x, y)}{\partial(r, \theta)}$ & $\frac{\partial(r, \theta)}{\partial(x, y)}$

$$\Rightarrow \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{vmatrix}$$

$$\frac{\partial x}{\partial r} = \cos \theta \quad ; \quad \frac{\partial y}{\partial r} = \sin \theta \quad ; \quad \frac{\partial x}{\partial \theta} = -r \sin \theta, \quad \frac{\partial y}{\partial \theta} = r \cos \theta$$

$$\frac{\partial(x,y)}{\partial(\theta,0)} = \begin{vmatrix} \partial x / \partial \theta & \partial x / \partial 0 \\ \partial y / \partial \theta & \partial y / \partial 0 \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$\frac{\partial(x,y)}{\partial(\theta,0)} \Rightarrow r \cos^2 \theta + r \sin^2 \theta \Rightarrow r$$

$$\frac{\partial(x,y)}{\partial(\theta,0)} \frac{\partial(\theta,0)}{\partial(x,y)} = 1$$

$$\frac{\partial(\theta,0)}{\partial(x,y)} = \frac{1}{r}$$

Q $u_1 = \frac{x_2 x_3}{x_1}$; $u_2 = \frac{x_1 x_3}{x_2}$; $u_3 = \frac{x_1 x_2}{x_3}$ find $J(u_1, u_2, u_3) = 1$

$$\Rightarrow \begin{vmatrix} \partial u_1 / \partial x_1 & \partial u_1 / \partial x_2 & \partial u_1 / \partial x_3 \\ \partial u_2 / \partial x_1 & \partial u_2 / \partial x_2 & \partial u_2 / \partial x_3 \\ \partial u_3 / \partial x_1 & \partial u_3 / \partial x_2 & \partial u_3 / \partial x_3 \end{vmatrix} \quad \begin{aligned} \frac{\partial u_1}{\partial x_1} &= x_2 x_3 \left(-\frac{1}{x_1^2} \right) \\ \frac{\partial u_2}{\partial x_1} &= \frac{x_3}{x_2} \quad ; \quad \frac{\partial u_3}{\partial x_1} = \frac{x_2}{x_3} \\ \frac{\partial u_1}{\partial x_2} &= \frac{x_3}{x_1} \quad ; \quad \frac{\partial u_2}{\partial x_2} = x_1 x_3 \left(-\frac{1}{x_2^2} \right) \end{aligned}$$

$$\Rightarrow \begin{vmatrix} -\frac{x_2 x_3}{x_1^2} & x_3/x_1 & x_2/x_1 \\ x_3/x_2 & -\frac{x_1 x_3}{x_2^2} & x_1/x_2 \\ x_2/x_3 & x_1/x_3 & -\frac{x_1 x_2}{x_3^2} \end{vmatrix} \quad \begin{aligned} \frac{\partial u_3}{\partial x_2} &= \frac{x_1}{x_3} \quad ; \quad \frac{\partial u_1}{\partial x_3} = \frac{x_2}{x_1} \\ \frac{\partial u_2}{\partial x_3} &= \frac{x_1}{x_2} \quad ; \quad \frac{\partial u_3}{\partial x_3} = x_1 x_2 \left(-\frac{1}{x_3^2} \right) \end{aligned}$$

$$\frac{1}{x_1 x_2 x_3} \begin{vmatrix} -x_2 x_3 / x_1 & x_3 & x_2 \\ x_3 & -\frac{x_1 x_3}{x_2} & x_1 \\ x_2 & x_1 & -\frac{x_1 x_2}{x_3} \end{vmatrix}$$

$$\frac{1}{x_1 x_2 x_3} \left(-\frac{x_2 x_3}{x_1} (x_1^2 - x_1^2) - x_3 (-x_1 x_2 - x_1 x_2) + x_2 (x_1 x_3 + x_1 x_3) \right)$$

$$\frac{1}{x_1 x_2 x_3} (0 + 2x_1 x_2 x_3 + 2x_1 x_2 x_3) = 4 \text{ Ans.}$$

Q. $u = x + y + z$
 $v = x - y + z$
 $w = x^2 + y^2 + z^2 - 2yz$
 prove that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$

$$\Rightarrow \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$\frac{\partial u}{\partial x} = 1, \frac{\partial u}{\partial y} = 1, \frac{\partial u}{\partial z} = 1$$

$$\frac{\partial v}{\partial x} = 1, \frac{\partial v}{\partial y} = -1, \frac{\partial v}{\partial z} = 1$$

$$\frac{\partial w}{\partial x} = 2x, \frac{\partial w}{\partial y} = 2y - 2z, \frac{\partial w}{\partial z} = 2z - 2y$$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2x & 2y - 2z & 2z - 2y \end{vmatrix} \Rightarrow \begin{vmatrix} 2 & 0 & 0 \\ 1 & -1 & 1 \\ 2x & 2y - 2z & 2z - 2y \end{vmatrix}$$

$$\Rightarrow 2(-2z + 2y - 2y + 2z) = 0$$

Hence proved.

Q. $u = \frac{y^2}{2x}, v = \frac{x^2 + y^2}{2x}$ find $\frac{\partial(u, v)}{\partial(x, y)}$

$$\Rightarrow \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \Rightarrow \begin{vmatrix} -\frac{y^2}{2x^2} & \frac{y}{x} \\ \frac{1}{2} - \frac{y^2}{2x^2} & \frac{y}{x} \end{vmatrix} \Rightarrow \frac{-y^3}{2x^3} - \frac{y}{2x} + \frac{y^3}{2x^3} + \frac{y}{2x}$$

$$\frac{\partial u}{\partial x} = -\frac{y^2}{2x^2}, \frac{\partial u}{\partial y} = \frac{y}{x}$$

$$\frac{\partial v}{\partial x} = \frac{1}{2} - \frac{y^2}{2x^2}, \frac{\partial v}{\partial y} = \frac{y}{x}$$

$$\boxed{\text{Ans } -\frac{y}{2x}}$$

Q. $u = \frac{1}{x}$, $v = \frac{x^2}{y}$, $w = x + y + zy^2$ find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$

$$\Rightarrow \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$\frac{\partial u}{\partial x} = -\frac{1}{x^2}, \frac{\partial u}{\partial y} = 0, \frac{\partial u}{\partial z} = 0$$

$$\frac{\partial v}{\partial x} = \frac{2x}{y}, \frac{\partial v}{\partial y} = -\frac{x^2}{y^2}, \frac{\partial v}{\partial z} = 0$$

$$\frac{\partial w}{\partial x} = 1, \frac{\partial w}{\partial y} = 1 + 2zy, \frac{\partial w}{\partial z} = y^2$$

$$\begin{vmatrix} -1/x^2 & 0 & 0 \\ 2x/y & -x^2/y^2 & 0 \\ 1 & 1+2yz & y^2 \end{vmatrix} \Rightarrow \frac{-1}{x^2} (-x^2) = 1$$

Q. $x = r \cos \theta \cos \phi$ show that $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = -r^2 \cos \theta$
 $y = r \cos \theta \sin \phi$
 $z = r \sin \theta$

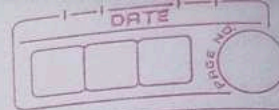
$$\Rightarrow \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix}$$

$$\frac{\partial x}{\partial r} = \cos \theta \cos \phi = x/r, \frac{\partial x}{\partial \theta} = -r \sin \theta \cos \phi = -z \cos \phi, \frac{\partial x}{\partial \phi} = -r \cos \theta \sin \phi = -y$$

$$\frac{\partial y}{\partial r} = \cos \theta \sin \phi = y/r, \frac{\partial y}{\partial \theta} = -r \sin \theta \sin \phi = -z \sin \phi, \frac{\partial y}{\partial \phi} = r \cos \theta \cos \phi = x$$

$$\frac{\partial z}{\partial r} = \sin \theta = z/r, \frac{\partial z}{\partial \theta} = r \cos \theta = x / \cos \phi, \frac{\partial z}{\partial \phi} = 0$$

$$f(x,y) = f(a,b) + [(x-a)f_x(a,b) + (y-b)f_y(a,b)] + \frac{1}{2!} [(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)] + \dots$$



* Taylor's series of two variable

$$f(a+h, b+k) = f(a,b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f + \frac{1}{2!} \left(h^2 \frac{\partial^2}{\partial x^2} + 2hk \frac{\partial^2}{\partial x \partial y} + k^2 \frac{\partial^2}{\partial y^2} \right) f + \frac{1}{3!} \left(h^3 \frac{\partial^3}{\partial x^3} + 3h^2 k \frac{\partial^3}{\partial x^2 \partial y} + 3h k^2 \frac{\partial^3}{\partial x \partial y^2} + k^3 \frac{\partial^3}{\partial y^3} \right) f + \dots$$

(ii) $f(x,y) = \cos x \cos y$ point $(0,0)$

$$f(a,b) = f(0,0) = 1$$

$$\frac{\partial}{\partial x} = -\sin x \cos y = f_x(0,0) = 0$$

$$\frac{\partial}{\partial y} = -\sin y \cos x = f_y(0,0) = 0$$

$$\frac{\partial^2}{\partial x^2} = -\cos x \cos y = f_{xx}(0,0) = -1$$

$$\frac{\partial^2}{\partial y^2} = -\cos y \cos x = f_{yy}(0,0) = -1$$

$$\frac{\partial^2}{\partial x^2 \partial y} = \cos x \sin y = f_{xxy}(0,0) = 0$$

$$\frac{\partial^2}{\partial y^2 \partial x} = \cos y \sin x = f_{yyx}(0,0) = 0$$

$$\frac{\partial^2}{\partial x \partial y} = \sin x \sin y = f_{xy}(0,0) = 0$$

$$\frac{\partial^3}{\partial x^3} = \sin x \cos y = f_{xxx}(0,0) = 0$$

$$\frac{\partial^3}{\partial y^3} = \sin y \cos x = f_{yyy}(0,0) = 0$$

Apply Taylor's theorem

$$f(x,y) = f(a,b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f + \frac{1}{2!} \left(h^2 \frac{\partial^2}{\partial x^2} + 2hk \frac{\partial^2}{\partial x \partial y} + k^2 \frac{\partial^2}{\partial y^2} \right) f + \frac{1}{3!} \left(h^3 \frac{\partial^3}{\partial x^3} + 3h^2 k \frac{\partial^3}{\partial x^2 \partial y} + 3h k^2 \frac{\partial^3}{\partial y^2 \partial x} + k^3 \frac{\partial^3}{\partial y^3} \right) f + \dots$$

$$f(x, y) = \frac{1}{0!} + (x x_0 + y y_0) f + \frac{1}{2!} (x^2 (-1) + 2xy x_0 + y^2 (-1)) f + \frac{1}{3!} (x^3 x_0 + 3x^2 y x_0 + 3y^2 x_0 + y^3 x_0) f$$

$$f(x, y) \Rightarrow 1 + \frac{1}{2!} (-x^2 - y^2) = 1 - \frac{(x^2 + y^2)}{2} \quad \text{Ans.}$$

iii) $f(x, y) = e^x \log(1+y)$

$$f(a, b) = f(0, 0) = 0$$

$$\frac{\partial}{\partial x} = e^x \log(1+y) = f_x(0, 0) = 0$$

$$\frac{\partial}{\partial y} = \frac{e^x}{1+y} = f_y(0, 0) = 1$$

$$\frac{\partial^2}{\partial x^2} = e^x \log(1+y) = f_{xx}(0, 0) = 0$$

$$\frac{\partial^2}{\partial y^2} = -\frac{e^x}{(1+y)^2} = f_{yy}(0, 0) = -1$$

$$\frac{\partial^2}{\partial x \partial y} = \frac{e^x}{1+y} = f_{xy}(0, 0) = 1$$

$$\frac{\partial^2}{\partial x^2 \partial y} = \frac{e^x}{1+y} = f_{xxy}(0, 0) = 1$$

$$\frac{\partial^2}{\partial y^2 \partial x} = -\frac{e^x}{(1+y)^2} = f_{yyx}(0, 0) = -1$$

$$\frac{\partial^3}{\partial x^3} = e^x \log(1+y) = f_{xxx}(0, 0) = 0$$

$$\frac{\partial^3}{\partial y^3} = -\frac{2e^x}{(1+y)^3} = f_{yyy}(0, 0) = 2$$

$$f(x, y) = f(0, 0) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f + \frac{1}{2!} \left(h^2 \frac{\partial^2}{\partial x^2} + 2hk \frac{\partial^2}{\partial x \partial y} + k^2 \frac{\partial^2}{\partial y^2} \right) f +$$

$$\frac{1}{3!} \left(h^3 \frac{\partial^3}{\partial x^3} + 3h^2 k \frac{\partial^3}{\partial x^2 \partial y} + 3k^2 h \frac{\partial^3}{\partial y^2 \partial x} + k^3 \frac{\partial^3}{\partial y^3} \right) f + \dots$$

$$\Rightarrow f(x,y) = 0 + (x \times 0 + y \times 1)f + \frac{1}{2!} (x^2 \times 0 + 2xy \times 1 + y^2 \times (-1))f + \frac{1}{3!} (x^3 \times 0 + 3x^2y \times 1 + 3y^2x \times (-1) + y^3 \times 2)f$$

$$f(x,y) = (y)f + \frac{1}{2} (2xy - y^2) + \frac{1}{3!} (3x^2y - 3xy^2 + 2y^3)f$$

Ans.

Q. $f(x,y) = e^x \cos y$ for point $(1, \pi/4)$

$$f_x(a,b) = e^x \cos y = f_x(1, \pi/4) = e \cos(\pi/4) = e/\sqrt{2}$$

$$f_y(a,b) = -e^x \sin y = f_y(1, \pi/4) = -e/\sqrt{2}$$

$$f_{xx}(a,b) = e^x \cos y = f_{xx}(1, \pi/4) = e/\sqrt{2}$$

$$f_{yy}(a,b) = -e^x \cos y = f_{yy}(1, \pi/4) = -e/\sqrt{2}$$

$$f_{xy}(a,b) = -e^x \sin y = f_{xy}(1, \pi/4) = -e/\sqrt{2}$$

$$\Rightarrow f(x,y) = f(a,b) + [(x-a)f_x(a,b) + (y-b)f_y(a,b)] + \frac{1}{2!} [(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)]$$

$$\Rightarrow f(x,y) = \frac{e}{\sqrt{2}} + \left[(x-1)\frac{e}{\sqrt{2}} - (y-\pi/4)\frac{e}{\sqrt{2}} \right] + \frac{1}{2!} \left[(x-1)^2 \frac{e}{\sqrt{2}} + 2(x-1)(y-\pi/4)\left(-\frac{e}{\sqrt{2}}\right) + (y-\pi/4)^2 \left(-\frac{e}{\sqrt{2}}\right) \right]$$

Ans.

maxima & minima of function of two variable

working rule -

let $f(x, y)$ be a function

1) find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$

2) put $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$ and find value of x & y

3) Calculate

$$r = \frac{\partial^2 f}{\partial x^2}, \quad s = \frac{\partial^2 f}{\partial x \partial y}, \quad t = \frac{\partial^2 f}{\partial y^2}$$

4) $rt - s^2 > 0$ and $r < 0$ funcⁿ maxima

5) $rt - s^2 > 0$ and $r > 0$ funcⁿ minima

6) $rt - s^2 < 0$ saddle point

7) $rt - s^2 = 0$ doubtful & further investigation.

Q. $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$

$$\frac{\partial f}{\partial x} = 4x^3 - 4x + 4y$$

$$\frac{\partial f}{\partial y} = 4y^3 + 4x - 4y$$

$$\frac{\partial f}{\partial x} = 0 \Rightarrow 4x^3 - 4x + 4y = 0 \Rightarrow x^3 - x + y = 0 \quad \text{--- (1)}$$
$$x(x^2 - 1) = -y \rightarrow \text{--- (1)}$$

$$\frac{\partial f}{\partial y} = 0 \Rightarrow 4y^3 + 4x - 4y = 0 \Rightarrow y^3 + x - y = 0 \quad \text{--- (2)}$$
$$y(y^2 - 1) = -x \quad \text{--- (2)}$$

eqⁿ (1) + 2

$$\boxed{x = -y}$$

$$x^3 - x - x = 0$$

$$x^3 - 2x = 0 \Rightarrow x(x^2 - 2) = 0$$

$$x^2 = 2x$$

$$\boxed{x = 0, \sqrt{2}, -\sqrt{2}}$$

$$y = 0, -\sqrt{2}, \sqrt{2}$$

LET STUDY HUB

* Lagrange's method of Undetermined multipliers working rule

let $f(x, y, z)$ be a given funcⁿ & $\phi(x, y, z) = 0$ if any condition

1) Lagrange's Auxiliary theorem (function)
 $F(x, y, z, \lambda)$, λ is Lagrange's multiplier

$$F(x, y, z, \lambda) = f(x, y, z) + \lambda \phi(x, y, z)$$

2) Obtains the equation

$$\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0 \quad \text{solving then and find } x, y, z \text{ in terms of } \lambda.$$

Q find the maximum & minimum distance from the point $(3, 4, 12)$ from the sphere $x^2 + y^2 + z^2 = 1$ (condition)

$$AP^2 = (x-3)^2 + (y-4)^2 + (z-12)^2 = f(x, y, z)$$

AP^2 = Distance between A & P

$$\Rightarrow \phi(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$$

$\Rightarrow$ Lagrange's Auxiliary function

$$F(x, y, z, \lambda) = f(x, y, z) + \lambda \phi(x, y, z)$$

$$= (x-3)^2 + (y-4)^2 + (z-12)^2 + \lambda (x^2 + y^2 + z^2 - 1)$$

$$\Rightarrow \frac{\partial F}{\partial x} = 2(x-3) + 0 + 0 + 2x\lambda = 0$$

$$2(x-3) + 2x\lambda = 0$$

$$2x - 6 + 2x\lambda = 0$$

$$2x(1+\lambda) = 6$$

$$1+\lambda = \frac{3}{x}$$

$$\Rightarrow \lambda = \frac{3}{x} - 1$$

$$x = \frac{3}{1+\lambda}$$

$$\frac{\partial F}{\partial y} = 2(y-4) + 2\lambda y = 0$$

$$2y - 8 + 2\lambda y = 0$$

$$1 + \lambda = \frac{8}{y \times 2}$$

$$\Rightarrow \boxed{y = \frac{4}{1+\lambda}}$$

$$\boxed{\lambda = \frac{4}{y} - 1}$$

$$\frac{\partial F}{\partial z} = 2(z-12) + 2\lambda z = 0$$

$$2z - 24 + 2\lambda z = 0$$

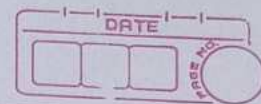
$$1 + \lambda = \frac{24}{2z}$$

$$\Rightarrow \boxed{z = \frac{12}{1+\lambda}}$$

$$\boxed{\lambda = \frac{12}{z} - 1}$$

$$\Rightarrow x^2 + y^2 + z^2 - 1 = 0$$

VECTOR DIFFERENTIATION



$$1) \frac{d}{dt} A \cdot B = A \frac{dB}{dt} + B \frac{dA}{dt}$$

$$2) \frac{d}{dt} (A \times B) = A \times \frac{dB}{dt} + \frac{dA}{dt} \times B$$

$$Q. A = 5t^2 \hat{i} + t \hat{j} - t^3 \hat{k}$$

$$B = \sin t \hat{i} - \cos t \hat{j}$$

$$\Rightarrow \frac{d}{dt} A \cdot B \Rightarrow A \frac{dB}{dt} + B \frac{dA}{dt}$$

$$= (5t^2 \hat{i} + t \hat{j} - t^3 \hat{k}) \cdot (\cos t \hat{i} + \sin t \hat{j}) + (\sin t \hat{i} - \cos t \hat{j}) \cdot (10t \hat{i} + \hat{j} - 3t^2 \hat{k})$$

$$= 5t^2 \cos t + t \sin t + 10t \sin t - \cos t$$

$$= 5t^2 \cos t - \cos t + 11t \sin t \quad \text{Ans.}$$

$$\Rightarrow \frac{d}{dt} A \times B \Rightarrow A \times \frac{dB}{dt} + \frac{dA}{dt} \times B \Rightarrow A \times (\cos t \hat{i} + \sin t \hat{j}) + (10t \hat{i} + \hat{j} - 3t^2 \hat{k}) \times B$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5t^2 & t & -t^3 \\ \cos t & \sin t & 0 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 10t & 1 & -3t^2 \\ \sin t & -\cos t & 0 \end{vmatrix}$$

$$\Rightarrow \hat{i} [0 + t^3 \sin t] - \hat{j} [0 + t^3 \cos t] + \hat{k} [5t^2 \sin t - t \cos t]$$

$$+ \hat{i} [0 - 3t^2 \cos t] - \hat{j} [0 + 3t^2 \sin t] + \hat{k} [-10t \cos t - \sin t]$$

$$\Rightarrow \hat{i} [t^3 \sin t - 3t^2 \cos t] - \hat{j} [t^3 \cos t + 3t^2 \sin t] + \hat{k} [5t^2 \sin t - t \cos t - 10t \cos t - \sin t]$$

Ans

* gradient of f (grad f or ∇f)
 The vector function ∇f is define as the gradient of scalar point function f and it written as grad f
 thus

$$\text{grad } f = \nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z}$$

$$\text{normal vector} = \text{grad } f = \nabla f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) f$$

Q $f = xy^3z^2 - 4$

$$\Rightarrow \text{normal vector} = \text{grad } f = \nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z}$$

$$\nabla f = i(y^3z^2) + j(xz^2 \times 3y^2) + k(xy^3 \times 2z)$$

$$\nabla f = y^3z^2 \hat{i} + 3xy^2z^2 \hat{j} + 2xy^3z \hat{k} \quad \text{Ans}$$

$$\Rightarrow \text{unit normal vector} = \frac{\text{grad } f}{|\text{grad } f|}$$

at point $(-1, -1, 2)$

$$= \frac{y^3z^2 \hat{i} + 3xy^2z^2 \hat{j} + 2xy^3z \hat{k}}{\sqrt{(y^3z^2)^2 + (3xy^2z^2)^2 + (2xy^3z)^2}}$$

$$= \frac{-4 \hat{i} - 12 \hat{j} + 4 \hat{k}}{\sqrt{(4)^2 + (12)^2 + (4)^2}}$$

$$= \frac{4(-\hat{i} - 3\hat{j} + \hat{k})}{4(\sqrt{1+9+1})}$$

$$= \boxed{\frac{-\hat{i} - 3\hat{j} + \hat{k}}{\sqrt{11}}} \quad \text{Ans.}$$

* Directional Derivative = $(\text{grad } f)_p \frac{\vec{N}}{|\vec{N}|}$

$\vec{N}$ = direction of vector

ex- $f(x, y, z) = xy^2 + yz^3$

at point $(2, -1, 1)$

direction of vector $\vec{N} = \hat{i} + 2\hat{j} + 2\hat{k}$

$$\begin{aligned} \Rightarrow (\text{grad } f)_p &= \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \\ &= \hat{i}(y^2) + \hat{j}(2xy + z^3) + \hat{k}(3z^2 y) \\ &\quad \hat{i}(-1)^2 + \hat{j}(2(2)(-1) + 1) + \hat{k}(3(1)^2(-1)) \\ &= \hat{i}(1) + \hat{j}(-4 + 1) + \hat{k}(-3) \\ &= \hat{i} - 3\hat{j} - 3\hat{k} \text{ Ans.} \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Directional Derivative} &= (\text{grad } f)_p \frac{\vec{N}}{|\vec{N}|} \\ &= \frac{(\hat{i} - 3\hat{j} - 3\hat{k}) \cdot (\hat{i} + 2\hat{j} + 2\hat{k})}{\sqrt{1+4+4}} \\ &= \frac{1 - 6 - 6}{3} \\ &= \frac{-11}{3} \text{ Ans.} \end{aligned}$$

* Divergence = The divergence of continuously differential vector point function F , it denoted by $[\text{div } F = \nabla \cdot F]$
it is define by equation

$$\nabla \cdot F = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot F$$

$$F = f\hat{i} + \phi\hat{j} + \psi\hat{k}$$

$$\nabla \cdot F = \frac{\partial f}{\partial x} + \frac{\partial \phi}{\partial y} + \frac{\partial \psi}{\partial z}$$

Q. $u = x^3 + y^3 + z^3 - 3xyz$

$$\Rightarrow F = \text{grad} f = \nabla f = \hat{i} \frac{\partial u}{\partial x} + \hat{j} \frac{\partial u}{\partial y} + \hat{k} \frac{\partial u}{\partial z}$$

$$= \hat{i}(3x^2 - 3yz) + \hat{j}(3y^2 - 3xz) + \hat{k}(3z^2 - 3xy)$$

$$\Rightarrow \text{div} F = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot F$$

$$= 3 \frac{\partial}{\partial x} x^2 + 3 \frac{\partial}{\partial y} y^2 + 3 \frac{\partial}{\partial z} z^2$$

$$= 6x + 6y + 6z \quad \text{Ans.}$$

* $\text{curl} F = \nabla \times F = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times F$